

PRACTICAL REPORT 03

FULL STACK DEVELOPMENT (FSD) — SEMESTER 5

Topic: Responsive Web Design & CSS Layouts

Source: Campus Companion Project Implementation

Practical Objectives & Tasks:

- Task I:** Create a responsive web page using the viewport meta tag in the document `<head>`.
- Task II:** Create a responsive web page using width and max-width properties, adapting images to browser width.
- Task III:** Demonstrate responsive text sizing using the vw (viewport width) unit in CSS with `clamp()`.
- Task IV:** Demonstrate CSS media queries for creating adaptive breakpoints across device screen sizes.

Core Principles of Responsive Web Design (RWD):

Responsive Web Design enables modern web applications to dynamically adapt their layout, typography, and media across a diverse range of device viewports (smartphones, tablets, laptops, and ultra-wide desktops) using a single, unified codebase.

TASK I Viewport Meta Tag in HTML Document Head

Technical Explanation:

By default, mobile browsers render desktop-optimized websites on a virtual canvas (typically 980px wide) and scale it down. The `<meta name="viewport" content="width=device-width, initial-scale=1.0">` directive instructs the mobile rendering engine to set the viewport width directly to the physical screen width (`width=device-width`) and maintain a 1:1 initial zoom scale (`initial-scale=1.0`), enabling accurate media query matching and fluid CSS scaling.

Project Source Code: frontend/index.html

```
<!doctype html>
<html lang="en">
  <head>
    <meta charset="UTF-8" />
    <link rel="icon" type="image/svg+xml" href="/favicon.svg" />

    <!-- === Task I: Viewport Meta Tag for Responsive Design ===
      The viewport meta tag ensures the page scales correctly on all devices.
      'width=device-width' sets the viewport width to the device's screen width.
      'initial-scale=1.0' sets the initial zoom level to 100%. -->
    <meta name="viewport" content="width=device-width, initial-scale=1.0" />

    <meta name="description" content="Campus Companion – Smart attendance tracking for college students." />
    <title>Campus Companion – Attendance Tracker</title>
  </head>
  <body>
    <div id="root"></div>
    <script type="module" src="/src/main.jsx"></script>
  </body>
</html>
```

TASK II Responsive Images and Containers (width & max-width)

Technical Explanation:

Using `max-width: 100%` alongside `height: auto` guarantees that media elements will never exceed the physical boundaries of their parent container while preserving their intrinsic aspect ratio. For cards and modals, setting `width: 100%` combined with a constraining `max-width` (e.g. `max-width: 480px`) allows containers to stretch on narrow mobile screens while remaining elegantly bounded on large desktops.

Project Source Code: `frontend/src/styles/index.css` & `frontend/src/layouts/AuthLayout/AuthLayout.module.css`

```
/* == Task II: Responsive Images in Global CSS (frontend/src/styles/index.css) == */
img {
  max-width: 100%;
  width: auto;
  height: auto;
  display: block;
}

.responsive-img {
  width: 100%;
  max-width: 100%;
  height: auto;
  object-fit: contain;
}

/* == Task II: Responsive Container Card (frontend/src/layouts/AuthLayout/AuthLayout.module.css) == */
.card {
  position: relative;
  z-index: 1;
  width: 100%;
  max-width: 480px; /* Constrained on desktop, 100% width on mobile */
  margin: var(--space-4);
}
```

TASK III Responsive Text Sizing using vw (Viewport Width) Units

Technical Explanation:

The vw unit represents 1% of the viewport width. Combining vw with CSS `clamp(MIN, PREFERRED_VW, MAX)` produces **fluid typography**: text scales smoothly in real time with window resizing without abrupt jumps, while staying safely clamped between accessible minimum and aesthetic maximum font thresholds.

Project Source Code: `frontend/src/styles/index.css` & `frontend/src/pages/DashboardPage/DashboardPage.module.css`

```
/* === Task III: Fluid Typography using clamp() and vw (frontend/src/styles/index.css) === */
h1 { font-size: clamp(1.75rem, 4vw, 2.25rem); } /* Scales fluidly from 28px to 36px */
h2 { font-size: clamp(1.375rem, 3vw, 1.875rem); } /* Scales fluidly from 22px to 30px */
h3 { font-size: clamp(1.125rem, 2.5vw, 1.5rem); } /* Scales fluidly from 18px to 24px */
h4 { font-size: clamp(1rem, 2vw, 1.25rem); } /* Scales fluidly from 16px to 20px */

/* Responsive large numerical statistics (Dashboard & Cards) */
.stat-number {
  font-size: clamp(1.5rem, 3.5vw, 2.5rem);
  font-family: var(--font-heading);
  font-weight: var(--weight-bold);
}

/* Responsive Stat Values in Dashboard (DashboardPage.module.css) */
.statValue {
  font-size: clamp(1.25rem, 3vw, 1.5rem);
  font-weight: var(--weight-bold);
  line-height: 1;
}

.sectionTitle {
  font-size: clamp(1.125rem, 2.5vw, 1.25rem);
  font-weight: var(--weight-semibold);
}
```

TASK IV Media Queries for Adaptive Breakpoints

Technical Explanation:

Media queries (`@media`) apply conditional CSS rules based on device characteristics such as viewport width. Our Campus Companion application utilizes standard architectural breakpoints (1200px, 1024px, 768px, 640px, 480px) to transition from multi-column grids to optimized stacked mobile layouts.

Project Source Code: frontend/src/pages/DashboardPage/DashboardPage.module.css & frontend/src/styles/index.css

```
/* — Task IV: Adaptive Media Query Breakpoints (DashboardPage.module.css) — */

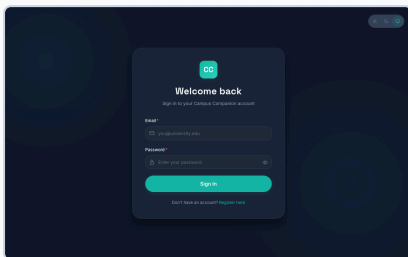
/* Tablet Breakpoint (<= 1024px): 4-column grid transitions to 2-column grid */
@media (max-width: 1024px) {
  .statsRow {
    grid-template-columns: repeat(2, 1fr);
  }
}

/* Mobile Breakpoint (<= 640px): 2-column grid collapses to 1-column single stack */
@media (max-width: 640px) {
  .statsRow {
    grid-template-columns: 1fr;
  }
  .subjectGrid {
    grid-template-columns: 1fr;
  }
  .subjectCard {
    flex-direction: column;
    align-items: center;
    text-align: center;
  }
}

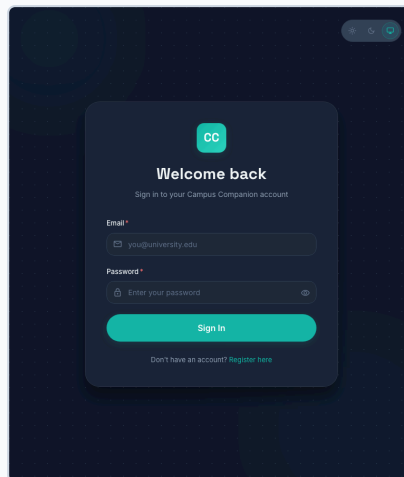
/* Mobile Breakpoint (<= 768px): Hide desktop sidebar and resize headings */
@media (max-width: 768px) {
  .hide-mobile {
    display: none !important;
  }
  h1 { font-size: clamp(1.5rem, 5vw, 2rem); }
}
```

VISUAL VERIFICATION

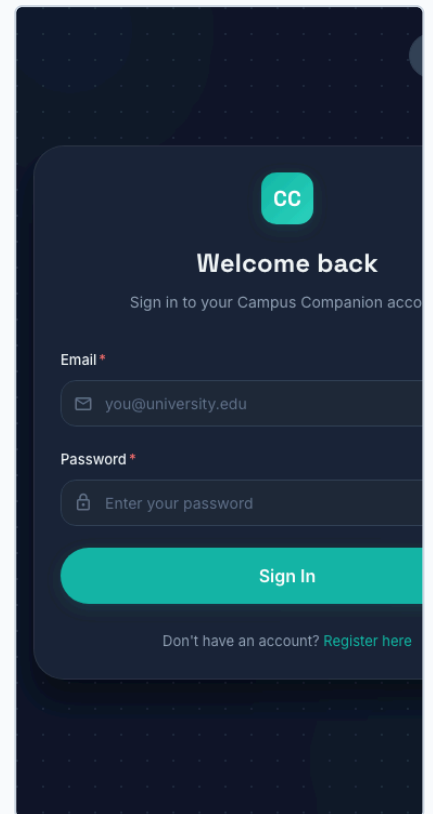
Campus Companion Multi-Device Responsiveness (Screenshots)

 Desktop View

Viewport: 1280 × 800px
(Max-width card & fluid orbs)

 Tablet View

Viewport: 768 × 900px
(Fluid typography scaling)

 Mobile View

Viewport: 375 × 750px
(100% width container adaptation)

Practical Conclusion & Learning Outcomes:

- Implemented the foundational HTML5 viewport meta tag to ensure 1:1 hardware pixel scaling on mobile devices.
- Applied defensive responsive container strategies using `width`, `max-width: 100%`, and `height: auto` for media.
- Employed CSS `vw` units coupled with `clamp()` for modern fluid typography across all heading and statistical elements.
- Constructed adaptive multi-tiered CSS media queries establishing clean breakpoints across desktop (1280px), tablet (768px), and mobile (375px) viewports in our **Campus Companion** web application.